

## **Group4:**

### **Team Members:**

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## **Project: Transport Management System (Project Description)**

A **Transport Management System (TMS)** is an small-level software solution designed to manage all transportation operations efficiently. It handles vehicle scheduling, route planning, driver management, fuel tracking, maintenance records, passenger or goods transport, and billing. The system is widely used in universities, companies, logistics firms, and public transport authorities to automate transport operations and improve efficiency.

## **1. Project Overview**

The main purpose of this system is to replace manual transport management with a digital platform that ensures:

- Efficient vehicle utilization
- Real-time route and schedule management
- Accurate tracking of drivers and vehicles
- Fuel consumption monitoring
- Maintenance scheduling
- Transparent billing system

It can be implemented as:

- Desktop Application (Java Swing or SWT GUI, Python, PyQt6) , Database MS SQL Only. **Notice, I don't want web base.**

## **2. User Roles**

### **Admin (Transport Manager)**

- Manage vehicles
- Assign routes

- Manage drivers
- View reports and analytics

## **Driver**

- View assigned route
- Update trip status
- Report vehicle issues

## **Passenger / Student / Employee**

- Request transport facility
- View routes and timings
- Track vehicle (if enabled)

## **Maintenance Staff**

- Vehicle inspection
- Repair scheduling
- Maintenance logs

# **3. Core Modules**

## **Vehicle Management**

- Add/update vehicles
- Vehicle type (bus, van, car)
- Registration details
- Capacity tracking

## **Route Management**

- Define routes (pickup → drop)
- Stop management
- Distance calculation

## **Driver Management**

- Driver profiles
- License verification
- Assignment history

## **Scheduling System**

- Daily transport schedule
- Pickup and drop timing
- Route-wise allocation

## **Fuel Management**

- Fuel consumption tracking
- Fuel refill records
- Cost calculation

## **Maintenance Module**

- Repair history
- Service schedule
- Breakdown reporting

## **Billing System**

- Monthly transport fees
- Due payments
- Fine management

## **Tracking System (Advanced)**

- Vehicle tracking
- Route deviation alerts

## **4. Database Entities (Tables)**

- vehicles
- drivers
- routes
- stops
- schedules
- passengers
- bookings
- payments
- fuel\_records
- maintenance\_records
- assignments
- transport\_fees

- tracking\_logs

## 5. Key Features

Real-time route scheduling  
Driver and vehicle assignment system  
Fuel consumption monitoring  
Maintenance alerts  
Monthly billing system  
Route optimization  
Tracking  
Automated reporting system

## 6. System Workflow

### Transport Assignment Process:

1. Admin creates route
2. Vehicle is assigned
3. Driver is assigned
4. Passengers are allocated
5. Schedule is generated
6. Transport runs daily operations

### Maintenance Process:

1. Driver reports issue
2. Maintenance request is created
3. Repair is scheduled
4. Vehicle status updated

### Payment Process:

1. Passenger selects transport service
2. Monthly fee is generated
3. Payment is recorded
4. Due balance is updated

## 7. System Interfaces

## **Passenger Panel**

- View routes
- Transport booking
- Payment history

## **Admin Dashboard**

- Full system control
- Reports and analytics
- Vehicle tracking

## **Driver Panel**

- Assigned routes
- Trip status updates

## **Maintenance Panel**

- Repair logs
- Vehicle condition reports

# **8. Security Features**

- Role-based login system
- Secure driver verification
- Encrypted financial data
- Audit logs for operations